The Rope Hypothesis: state of the programme

An account for a reader outside the project. Core hypothesis due to Bill Gaede; developed and formalised in this corpus.

518 registered claims, 497 code-backed and passing, 111 Derived, 25 registered Failed and kept.

In one line: one medium of strands under tension reproduces optics, electricity, magnetism, local gravity and the nuclear mass table with the same constants throughout — and commits to a nineteen-entry inventory of falsifiable predictions, one of which is distinctive from standard physics and testable this decade.

(The previous edition, 17 July 2026, is archived at `history/_archive_STATE_2026-07-17.md`. It is superseded several times over: a large audit campaign on 1 August retired one sector and demoted most of the prediction list; the strong-field campaign of 2 August carried the gravity sector from its oldest assumption to the Hawking law-form; and the work of 5–7 August computed the horizon's "quantum of the ring", surfaced a striking cross-sector action coincidence, consolidated the electron/ħ and alpha results and the framework's own primitives into dedicated papers, and fixed the working mesh point. The sections below reflect all of it.)

What the framework says

Space is a material medium made of strands under tension. Matter is knotted or clasped structure in that medium; light is a transverse disturbance travelling through it; gravity is the effective metric such disturbances experience. One medium, one tension, no fields laid on top.

That is an old idea and by itself it is not physics. What distinguishes this corpus from previous statements of it is that every claim is a computation with a pre-registered pass criterion, and failures stay in the registry. The result is a programme that can be — and repeatedly has been — refuted by its own instruments.

Scope: ordinary matter (first generation and their composites), the photon, emergent relativity, and a gravity sector. It claims nothing about heavier generations, and lepton masses are registered as irreducible inputs (PM-004, a kept negative).

What the framework does

Two things, and they should be judged separately: it **reconstructs** classical physics from one ontology, and it **predicts** a falsifiable inventory beyond it whose *distinctive-from-standard-physics* content is narrow. The first is substantial and is the reason to read further. The second is real but easy to mis-count in either direction, so it is laid out carefully below: nineteen falsifiable predictions are on the books, eight of them distinctive in observable outcome, and exactly one currently distinctive *and* checkable *and* live. The honest headline is neither "almost nothing" nor "nineteen ways to tell it apart" — it is that the framework commits to nineteen testable relations while conceding that most overlap observationally with standard physics today.

What the framework possesses

Three quantities anchor everything downstream:

- **Flux-tube tension** T = 1.878e5 J/m — a hadronic measurement.
- **Strand spacing** $w = a/\sqrt{3}$, fixed *exactly* relative to the mesh scale a by an invariance theorem (ELEC-053): $T_0/\Sigma = a^2/3$ for any tube radius, so the spacing never depended on the disputed quantities it was long thought to. The absolute mesh scale is now measurement-fixed at the M-point (a = 6.0e-17 m, FND-MATTER-044), the m_e-consistent point that sits at 60% of the Lorentz bound rather than on it.
- **Vacuum stiffness** Σ — the one genuinely open number, with two registered candidates 28% apart: the framework's 5.1e35 J/m^3 and a lattice-anchored 3.61e35 computed from published QCD flux-tube data.

The classical sectors work, and this is the programme's real substance. It deserves stating before the prediction accounting, because a reader who meets only the prediction list will badly misjudge what the framework has done.

Light — 10 of 10 optical phenomena derived (OPT-001..010, all *Derived*): Huygens construction, single-slit diffraction, cavity standing waves, waveguide TE_{10} cutoff, total internal reflection, Snell's law from wavefront continuity, Fresnel coefficients with energy conservation, the Brewster angle with polarization as transverse-mode structure, and multilayer optics (AR coatings, Fabry–Pérot, dielectric mirrors). The medium's impedance $Z = \sqrt{(T\mu)} = T/c$ is intrinsic — not fitted.

Electricity. Charge is the topological linking number, so quantisation is not imposed but forced (EM-001). Current is transported linking, with no-wind-up, continuity $\nabla \cdot J = 0$ and the closed-loop condition shown to be one condition rather than three (EM-008), and a concrete screw mechanism that transports linking and delivers power (EM-014). The electrostatic sign theorem is derived, not assumed: like windings repel, opposite attract, with the 1/d Coulomb form, because winding is a constraint source (EM-015).

Magnetism works with no free parameters and no new predictions. The magnetic force between currents comes out of the swinging-rope field energy with the correct magnitude, the correct 1/d falloff and the correct signs (EM-012); the Lorentz force $F = qv\times B$ follows from the gauge-forced coupling, automatically perpendicular to both and doing no work (EM-013). Maxwell's equations follow from Bianchi identities plus Chern–Weil in d = 3 (EM-003), and the structural constants come with them: $Z_0 = \sqrt{(\mu_0/\varepsilon_0)} = 376.74\ \Omega$ and $c^2 = 1/(\mu_0\varepsilon_0)$ (EM-002).

Local gravity. The four classical weak-field tests take their GR values — light deflection 1.751″ against the photographed 1.75″, Mercury's 43.0″/century, Shapiro $\gamma = 1$, Nordtvedt $\eta = 0$ (GRV-002, *Derived*). The old caveat that this followed only *given* a matched metric is retired: the classical response channels were ruled out by a no-go theorem ($\gamma \in [-1, 0]$, kept at full strength), and the metric is now carried by the quantum-induced channel — the IR-universal remainder of the medium's induced action is Einstein–Hilbert (GRV-025, *Derived*), the physical one-metric derivation makes the identification and the full classical table unconditional (GRV-026/028/029, *Derived*), the strong-field extrapolation is certified a controlled expansion (GRV-048), and the induced Newton constant carries a derived exponent pair with $\zeta = 1.208$ (GRV-

075). The Poisson equation and the 1/r conditioning field are recovered from network statics (GRV-005), and the MOND acceleration scale emerges as $g† = cH_0/2π$ at zero free parameters, matching SPARC rotation curves.

Frame dragging — the structure is there, the magnitude is not yet derivable. The framework's twist sector supplies a gravitomagnetic mode with exactly the Lense-Thirring form: a locking mass term is symmetry-forbidden (kappa = 0 by Goldstone's theorem applied to the derived internal-azimuth breaking, GRV-067), the resulting mode is massless for any modulus (GRV-068), and the far field comes out with the $1/r^2$ falloff and dipole angular structure Lense-Thirring requires, from a Poisson equation plus a derived lemma with no free parameter (GRV-066). That structural result is genuine. What it cannot yet do — as a reviewer-prompted audit established (GRV-069..071) — is put a *number* on the effect: the registered strand action lacks a matter-to-rotation coupling and a metric shift-slot, so four of the five coefficients needed for the amplitude are absent. The honest status is therefore precise: the framework derives the *form* of frame dragging and matches Gravity Probe B's structure, but "can predict neither 37.2 mas/yr nor its absence and should claim neither" (GRV-071). The original source-audit failure that opened this line (GRV-059) is kept; the six sessions since converted it from "the sector is missing" to "the sector's structure is derived, its magnitude is the open frontier."

Nuclear masses across the table. Atomic masses are *predicted* from carbon-12 to uranium-238 with **one calibrated nuclear constant** (NUC-005, amended by NUC-018): bond counting from mode overlap, with the surface term derived from geometry and the Coulomb coefficient derived rather than fitted. Accuracy on mass is 0.00–0.51% across that range. For comparison, the standard semi-empirical mass formula uses roughly five fitted parameters to do the same job. Stated with its warts: the parameter-free surface ratio came out at 1.34 against 1.130 empirical (NUC-016 corrected an earlier 1.108 that had looked better for the wrong reason), the volume coefficient sits 12.8% high after that correction, and the derived Coulomb term is 9% off. The corrections *degraded* the headline and were registered anyway.

Beyond these: Born's half-angle law from classical rope wave energetics, bond mechanisms and molecular geometry (bent H_2O derived, H_2S at 92° supporting), the Yukawa form exact from a screened-mode Green's function, and a quantum foundations programme that, in the ribbon-production arc of August 2026 (QB-028..033), derived an end-to-end Bell violation from nucleation alone — CHSH 2.04 as a worst-case floor rising to 2.23 under the forced transport law, every number traced to three medium parameters, with verified no-signalling.

That one ontology yields light, electricity, magnetism, local gravity and the nuclear mass table together — from strands under tension, with the same constants throughout — is the substantive claim of this programme. By sector count: 15 of 34 EM claims are *Derived*, and 27 of 101 gravity claims — the gravity sector having more than doubled through the 2026 audit and strong-field campaigns, most of the additions Modeled with locked bars and code.

On predictive equivalence, and why it is not a concession. Most of the above reproduces results standard physics also produces, and no measurement separates two frameworks that agree on every observable. That is a fact about the situation, not a verdict on it. The case for this ontology rests on explanatory economy: one medium, one tension, the same constants across optics, electricity, magnetism and gravity, with no fields laid on top and no separate postulate per phenomenon. Snell's law from wavefront continuity in a medium is a different kind of statement from Snell's law as a boundary condition on a field. Whether that difference matters is a judgement about what a physical explanation is for — and the history is not one-sided: Lorentzian and Einsteinian relativity were predictively identical for years, and the argument between them was ontological.

Two things this document will not claim. First, that predictive equivalence makes the standard accounts mere curve-fitting. They are not: QED predicted the electron anomalous moment to twelve digits before it was measured, and antimatter, the W and Z masses, and gravitational waves were all committed to in advance and found. That is real, and a framework that repeatedly sticks its neck out and survives has earned something this corpus has not yet earned. But the comparison must be read with the parameter counts in view, or it flatters the wrong side. The Standard Model buys its accuracy with roughly nineteen free parameters fitted to data — the fermion masses, the mixing angles, the couplings, the Higgs sector — none derived from a deeper principle. This framework is trying to reach the same physics from a handful of primitives, and where it succeeds it does so at *zero* or *one* free parameter (the nuclear table on a single calibration; the SPARC acceleration relation at zero, GRV-030). Higher precision purchased with nineteen knobs is a different achievement from lower precision derived from almost none, and a fair reading says so. This document concedes the Standard Model's predictive record without conceding that the record settles which account explains more from less. Second, that this framework explains mass in general. The distinction matters and cuts both ways. COMPOSITE masses it does well: the nuclear table from the light nuclei through uranium falls out of bond counting with one calibration, and the semi-empirical mass formula is now derived from that geometry across all five terms rather than fitted. FUNDAMENTAL masses and constants it does not: the lepton spectrum is registered as irreducible input (PM-004, a kept negative), $G \sim 1/(Ta)$ is assumed rather than derived with PRED-003 provisional on it, and the vacuum stiffness Σ remains free with two candidate values — a short list of unknowns against the Standard Model's longer one.

That is worth comparing squarely with the field picture, which is often credited with explaining mass and does not. The Higgs mechanism explains how mass is *compatible with gauge symmetry* — real work, and it correctly predicted the W/Z mass ratio — but every fermion mass is a free Yukawa coupling, fitted, spanning six orders of magnitude with no principle selecting them. Neither framework derives the fundamental masses. This one derives more of the composite ones, from a mechanism, with fewer knobs.

The prediction accounting that follows therefore asks a narrower question — where does this framework say something *different* — and should be read as a supplement to the reconstruction above, not a verdict on it.

What the framework predicts, testably

Nineteen falsifiable predictions; one that is distinctive, checkable, and live today. The full inventory is maintained in `papers/falsifiable_predictions.pdf` — each a stated mathematical relation with its inputs classified derived or measured, its test named, and its falsifier spelled out. A discrimination audit (1 August 2026) asked the strictest question of each: does its *observable outcome* differ from standard physics, is it checkable, and is it live? Exactly one passes all three (the α–G covariation below), which is why earlier editions led with "one prediction." But that strict filter is not the whole story: the corrected distinctiveness census counts **eight** predictions distinctive in observable outcome (ELEC-063), and **five** are testable with existing apparatus now — the α–G ratio, the heavy-hydride 90° asymptote (H_2Po), the detector nucleation-kinetics package, the Born exponent $p = 2$, and the analog cliff/coverage universality. The list also carries structural consequences (the Tsirelson cap $2\sqrt{2}$ named with its failed organ; the Koide gauge-boson structure), a confirmed-but-shared entry (cosmic birefringence), and a sharp retrodiction (the neutrino sum). The reader should not collapse this to "one" — one is the count that survives the *hardest* filter, not the count of predictions.

The distinctive-and-live one — PRED-003: $d \ln \alpha = -2\, d \ln G$. Both couplings are expressed in the same medium primitives, and cosmological drift running through the tension channel ties their rates by exactly −2. Drift through the strand spacing would give +1, so the prediction identifies a *channel*, not merely a number.

It has been confronted against published data. Four independent α-drift and three independent G-drift determinations combine to $\dot{\alpha}/\alpha = 9.9(1.1)\text{e}{-19}$ /yr and $\dot{G}/G = 1.2(0.7)\text{e}{-13}$ /yr; the relation holds at **1.74σ**. This is a null-versus-null consistency — survival, not confirmation.

Its substance is on the inverse side: the framework predicts $\dot{G}/G = -5.0(5.5)\text{e}{-19}$ /yr, five orders below current reach. **Any firm detection of $|\dot{G}/G|$ above 1.6e−18 /yr refutes it.** The named weak point is PSR J1713+0747, whose 2025 analysis gives +3.2(1.6)e−13 — the only nonzero-leaning G determination in the set, and 2.06σ against the relation on its own. A 3σ confirmation of that central value kills the claim and needs only a factor 1.45 in σ: reachable around **2027–2030** on baseline growth alone.

That is the corpus's **sharpest single bet** — the one prediction that is distinctive, checkable, and live at once. It is not the whole of what the framework wagers: the eight distinctive predictions and the five available-now tests above are also on the table, each with its own falsifier, several of them testable today. What makes PRED-003 the headline is that it is the one whose falsifier is both distinctive from standard physics and scheduled to fire this decade.

What was retired, and why

A census against four criteria — quantitative, distinctive in *observable outcome*, checkable, live — began at eight candidates and ended at one. The demotions are the honest content of the programme.

The mesoscopic-ħ picture is retired after six independent closures. It held that ħ's action length is a mesoscopic patch of the medium, implying Born statistics should fail near the nuclear scale — the corpus's one claim modified gravity could not make. It was checked and it failed, then survived nothing: nuclear data excluded it; the **Lorentz bound excluded it unaided**, requiring a strand scale 6.75× larger than the bound permits; sweeping the one free length over its entire allowed range showed the electron and ħ requirements pass **5.2 decades** apart; bundling, coherent-unit width, and a two-medium split each failed to close the gap. Five of the six needed no new experiment. The standing-wave *form* $S = \pi T A^2/2c$ survives untouched — the length cancels identically — and what selects its amplitude remains open.

Cosmic birefringence is confirmed but shared. The framework predicts EB/EE = $\sin(4\beta)/2$, flat in multipole, and Planck legacy data favours the constant model by Bayesian evidence across all four solutions. But the standard ultra-light axion predicts a constant angle too. The observable that *would* discriminate — frequency scaling — was derived and closes the door: material optical activity gives $\beta \propto \nu^2$, excluded at 4.9σ and overshooting the observed angle by 7e18 at the framework's own strand scale. Only the axion's ν^0 survives, and it must be imported as a postulate.

The neutrino sum $\Sigma m\nu$ = 58.47 meV is sharply falsifiable but sits where any minimal-normal-ordering model lands, so a confirmation would not select it.

The black-hole whisper is the nearest miss, and since the strong-field campaign it is also the site of the corpus's most interesting open dispute. Isolated holes go quiet instead of evaporating; fed holes emit reconnection noise whose frequency law — proportional to the surface gravity κ — is now derived by TWO independent routes: the original mode analysis ($\omega = 0.23\kappa$, GRV-040) and the new mechanism chain (statics → lift-over barrier → measured bit-cost → counted modes → Tolman redshift, with the depth cancelling exactly: $T_\infty \propto \kappa$, GRV-087). The two routes agree on the law, and the four-order coefficient disagreement registered between them (GRV-088) has since been ADJUDICATED IN-HOUSE (GRV-089..091): a category error — a temperature coefficient set against a ringing-frequency coefficient. The emission is impulsive (discrete reconnection snaps ring the throat at its resonance regardless of rate — Campbell's theorem), so the pitch is geometric and stands at 0.23κ while the mechanism's temperature sets the event rate. Both routes vindicated at their own questions; the black hole is a crackling-noise-class emitter — cold thermodynamics, bright snaps. The emission is genuinely sourced in the gravitational channel (a reconnection is purely deviatoric, 93% of excited power in the Einstein–Hilbert channel), but the strain is 7.9e−27 at 10 kpc, the best real candidate falls **36× short** of LIGO, and the corpus's own spectral work established a broadband quasi-thermal spectrum, removing four orders of line-search sensitivity. No detection claim is made in either direction of the coefficient dispute.

The frontier the earlier edition named as still open — the "quantum of the ring", the ħ scale at the horizon's door — has since been reached. A fourth exact cancellation (the campaign's signature move) gives a snap action A* = e_bit/ω_loc that is **independent of depth** (GRV-092), a genuine mass-independent quantum of the reconnection event. And a cross-registry action audit turned up the corpus's most suggestive coincidence to date (GRV-093): the horizon's emergent action and electromagnetism's fine-structure constant are written in the *same form* — a tension times an area over c — carried by two sectors that were never coupled. It is registered as a coincidence, not a derivation, but it is the kind of thing the programme exists to notice.

The physical Aharonov–Bohm branch is closed. Nothing in the corpus sources a flux, and the one topological circulation it derives ($2\pi N$) is exactly spectrally trivial in a 2π-periodic instrument.

The strong-field campaign (2 August 2026, GRV-074..088)

Twenty-one claims across three releases in a single day took the gravity sector from its oldest unexamined assumption to the Hawking law-form, with bars locked before every computation.

The assumed constitutive form of G was found dimensionally open, and the repair produced the corpus's first derived G exponent pair — the Sakharov route, with a 96-site lattice zero-point coefficient selecting a spacing of eight Planck lengths unprompted (GRV-075); a survey then found ZERO live pins of the lattice scale, reducing a two-scale tension to a single named fork with internal discriminators (GRV-076). The horizon's statics were derived rather than assumed: the support load is transverse-only, exactly what pressing supplies (GRV-077); the frozen star froze itself, every exterior disturbance dying away (GRV-078/079); and the exterior cannot be talked into danger — infalling focusing buys a factor of two before horizon-side removal wins (GRV-080). An energy-honest interior was then built after its predecessor was disqualified by its own audit: the ratchet-wave coupling, in which accretion emerged unasked (every infalling wave blueshifts past the local threshold and is partially eaten) and collapse became recorded structure — a permanent footprint of broken crossings with balanced books (GRV-081). The temperature chain followed link by labelled link: the bit-cost is 15.67× the barrier, exact to three digits across a factor-four sweep (GRV-082); the barrier is pressing × thickness for any lift path, by the fundamental theorem of calculus, no friction law needed (GRV-083); the conversion to temperature survived the refutation of its own first closure — fifteen barriers per bit cannot live in a two-level system — and settled on a mode count of 4–6, sourced one line each from three different sectors (GRV-084/085/086). Three exact cancellations then delivered the summit: the barrier-to-temperature ratio, the emission Boltzmann factor, and finally the observed temperature itself are all independent of depth, and $T_\infty = (\beta K h/m^*)\kappa$ — the Hawking law-form, with Hawking nowhere in the inputs (GRV-087). The coefficient evaluation returned the honest ending: four orders below the mode-analysis commitment, registered at full strength with four pre-listed suspects (GRV-088). The law is derived, and the number's court case is CLOSED: the coefficient campaign (GRV-089..091) promoted one suspect, eliminated one by exact cancellation, vindicated the mode identification by Campbell's theorem, and convicted the confrontation itself of a category error. Both of the corpus's routes stand at their own questions, and the item that arc left open — the quantum of the ring at the horizon's door — has since been computed: a depth-independent snap action (GRV-092), which in turn surfaced the same-form coincidence between the horizon's action and the fine-structure constant (GRV-093). The tensor coefficient and the induced-EH fork were carried further still (GRV-094/095).

What is weakest

The **electron sector** is the largest by claim count and contributes nothing at any prediction tier. Its central geometry — a clasp of two strands, sharply asymmetric and nearly planar — is stable and distinctive, but its charge form factor misses by four orders of magnitude, and that has been shown *not* to be a calibration error: shrinking the strand scale rescues the electron and breaks everything else. What the sector *did* yield, and what the electron/ħ paper now documents, is structural rather than predictive: the 3D radial profile was solved exactly (a cusped, scale-free, finite-energy family, ELEC-073), the framework's own no-material-points commitment selects the pure-tension functional rather than leaving it a choice (ELEC-077), the size comes out near 0.407 fm surviving two independent estimators (ELEC-081), and the spacing $w = a/\sqrt{3}$ is exactly invariant (ELEC-053). These are real results; none of them is a prediction that separates the framework from standard physics, which is why the sector is filed here.

The **pilot-wave branch** and the **gauge instrument** are both validated machinery with no target — correct mathematics, nil empirical content.

Two branches were retired on the same pattern: excellent instrument work built before anyone asked what would *source* the quantity being measured. That is recorded as a standing rule in this repository, and it cost roughly eight sessions to learn.

Method, and how to check any of this

Every claim carries an ID, a status (Derived / Modeled / EFT-constrained / Conjecture / Open / Failed) and where possible a benchmark recomputing its result from stated inputs.

- `claims.yaml` — the single source of truth.
- `tools/verify_corpus.py` — runs every benchmark; guards against registry corruption, empty parses, and parser drift.
- `analysis/*_bars_LOCKED.md` — pass criteria, written *before* the corresponding computation ran.
- `docs/technical/HBAR_SECTOR_CLOSURE.md` — the retirement record.
- `docs/technical/STANDING_RULE_SOURCE_BEFORE_INSTRUMENT.md` — the lesson from two closed branches.
- `docs/DETECTOR_KINETICS_PLAIN_LANGUAGE.md` — the strand-kinetics campaign (FND-STRAND-009..021) in plain language: what small isolated detectors do, the two-regime dark-count picture, the Derived power law, and the switch-on resolution.
- `papers/rope_foundations.pdf` — the medium's primitives and framework commitments, consolidated.
- `papers/rope_electron_hbar.pdf` and `papers/rope_alpha_paper.pdf` — the standing electron/ħ results and the fine-structure chain ($1/\alpha = 4\pi^3 D\_E$).
- `docs/ROPE_PARAMETERS.md` — the canonical parameter card (self-verified), with the M-point as the working mesh point.

Twenty-five claims are registered **Failed** and kept, including one theorem falsified by the corpus's own growth, an instrument disqualified by its own energy audit, and pre-committed bars that failed and were kept as the findings they were.

The failures are the point. A framework that recorded only its successes would have a longer list of predictions and no way to tell which of them meant anything.